

Ghanashyam Bharadwaz Avula

Hyderabad | Bengaluru | +91 7032975357 | bharadwaz004@gmail.com
linkedin.com/in/bharadwazavula | github.com/Bharadwaz004 | [Portfolio](#) | [Leetcode](#)

SUMMARY

Backend and AI Engineer who ships production-grade systems, combining software engineering with agentic workflows. Strong foundation in designing scalable APIs, AI agents, distributed systems and evaluation-driven applications with measurable performance improvements. Certified in Generative AI and machine learning by Anthropic and IBM. Seeking backend or AI engineering roles.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS

Backend & Web: FastAPI, Node.js, React, REST APIs, Microservices

Databases: PostgreSQL, MongoDB, Redis, ChromaDB

CI/CD & Cloud: Git, Docker, AWS, GCP

Gen AI: LangChain, Agentic AI, RAG, MCP, LLM Orchestration, LoRA Fine-Tuning, LLM Evaluation

ML: PyTorch, TensorFlow, Scikit-learn, PyTest

PROJECTS

Ecom Support Assistant | *Python, FastAPI, MCP, RAG, PostgreSQL, Evals*

[\[Live Link\]](#)

- Deployed a production grade e-commerce customer support assistant that answers queries grounded in store policy docs and a live orders database, exposed as an MCP server with a glass-box UI that streams the tool-call trace in real time.
- Designed a custom Model Context Protocol (MCP) architecture integrating multiple tools like RAG, order status retrieval and summarization with coordinated orchestration for seamless multi-tool execution.
- Validated on a 34-case eval harness of answerable, adversarial, and undocumented-topic queries: 33/34 answers fully grounded in retrieved context (LLM as judge), 97% retrieval recall and every out-of-scope query declined with zero false refusals.

AskMyDB - NL to SQL Engine | *LangChain, Node.js, Llama LLM, Graph RAG, SQLite, LoRA*

[\[Live Link\]](#)

- Orchestrated a natural language to SQL pipeline with hybrid retrieval (Vector + Graph RAG) for inter-table relations, multi-layer safety guardrails, and zero-config database onboarding for plain English to optimized SQL conversion.
- Engineered a LangChain pipeline using Llama LLM with Node.js middleware for JWT authentication, conversation memory, and robust guardrails for write-operation interception, rate limiting, and destructive query prevention, with 100% success across 31 functional test cases including complex multi-table joins.
- Evaluated the performance of LoRA finetuned Llama 8B model, which cut latency by 83% and cost by 92% vs. the 70B baseline model on a fixed schema, but kept the frontier model as default after it performed better on unseen databases.

Distributed Rate Limiter | *REST API, FastAPI, Redis, Lua, Docker, Nginx*

[\[Github\]](#)

- Built a Redis-backed distributed rate-limiting middleware for FastAPI that enforces exact per-key request limits across multiple load-balanced replicas, with a live dashboard to fire request bursts and watch limiter decisions in real time.
- Implemented a token bucket algorithm as an atomic Redis Lua script, eliminating read-check-write race conditions, with configurable fail-open/fail-closed modes on a Dockerized Nginx and 3-replica stack.
- Proved exactness under load with a zero-tolerance concurrency test admitting exactly 100 of 2,000 concurrent requests across replicas, adding only ~0.8ms latency overhead while sustaining 4,000+ req/s throughput via Nginx.

EXPERIENCE

AI-ML Virtual Intern

Google / AICTE | Remote | 04/2024 - 08/2024

- Architected a CNN model in TensorFlow/Keras for breast cancer detection, raising F1 from 0.78 to 0.92 across a 2,000-image dataset.

EDUCATION

Sreenidhi Institute of Science and Technology | Hyderabad, India

Bachelor of Technology in Computer Engineering | 08/2022 - 05/2026 | CGPA: 7.8/10

CERTIFICATIONS & ACHIEVEMENTS

- Participated in Smart India Hackathon 2023 as a backend developer, collaborating with my team on “Smart Crop Management System for Farmers” project.
- Claude code using Google Cloud Vertex AI - Anthropic
- IBM Machine Learning Specialization - Coursera
- Secured 97 percentile in GATE 2026.